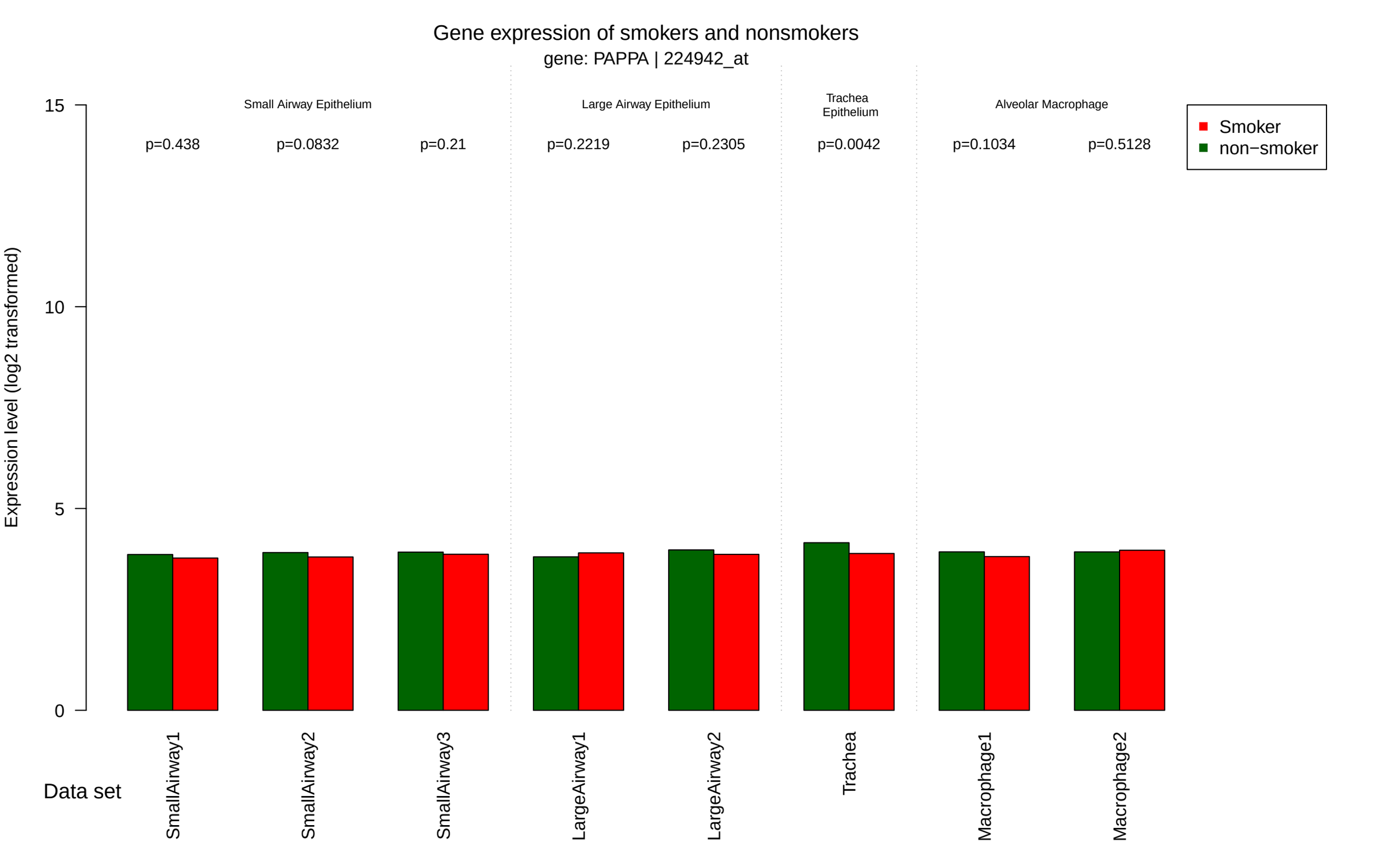
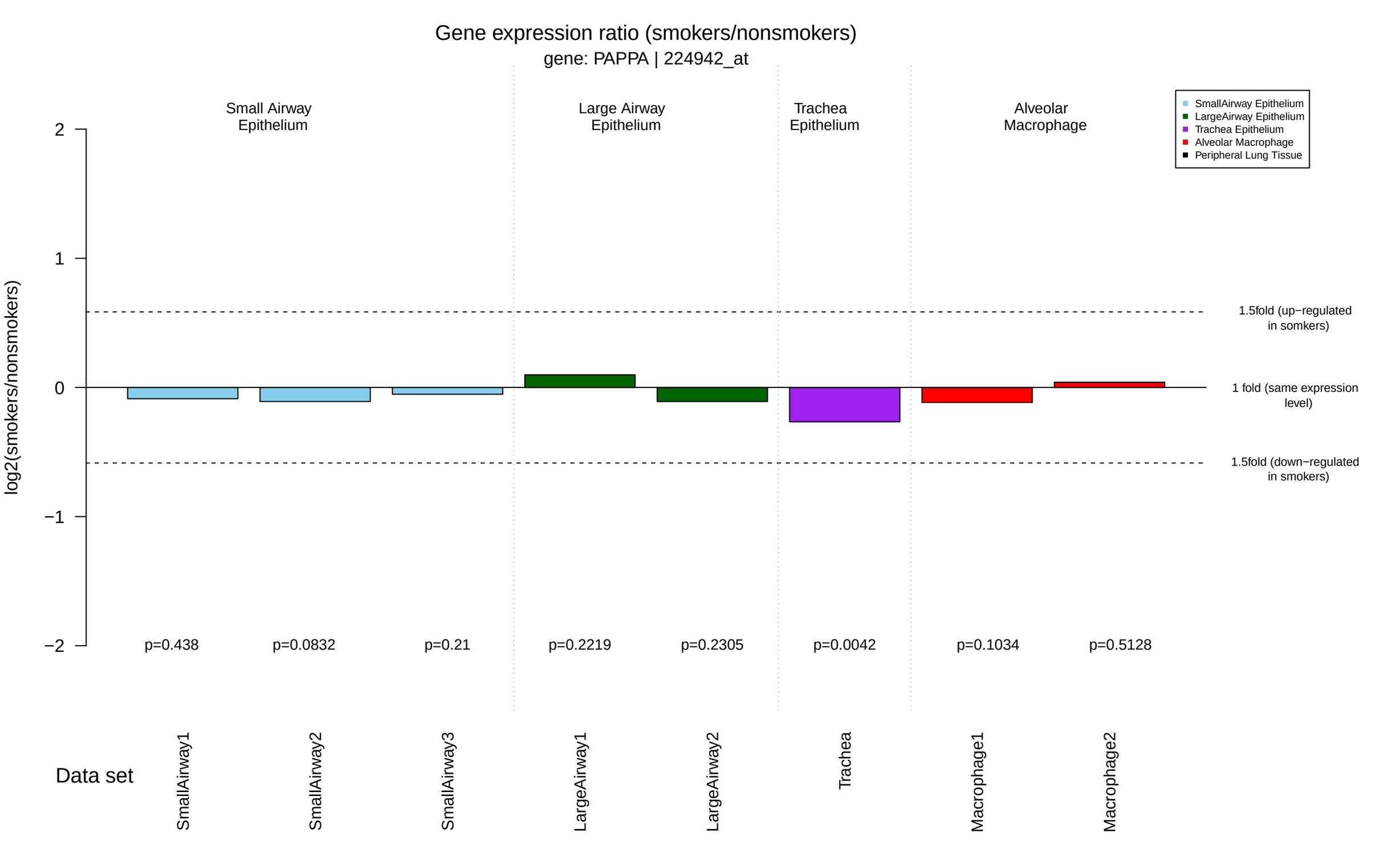
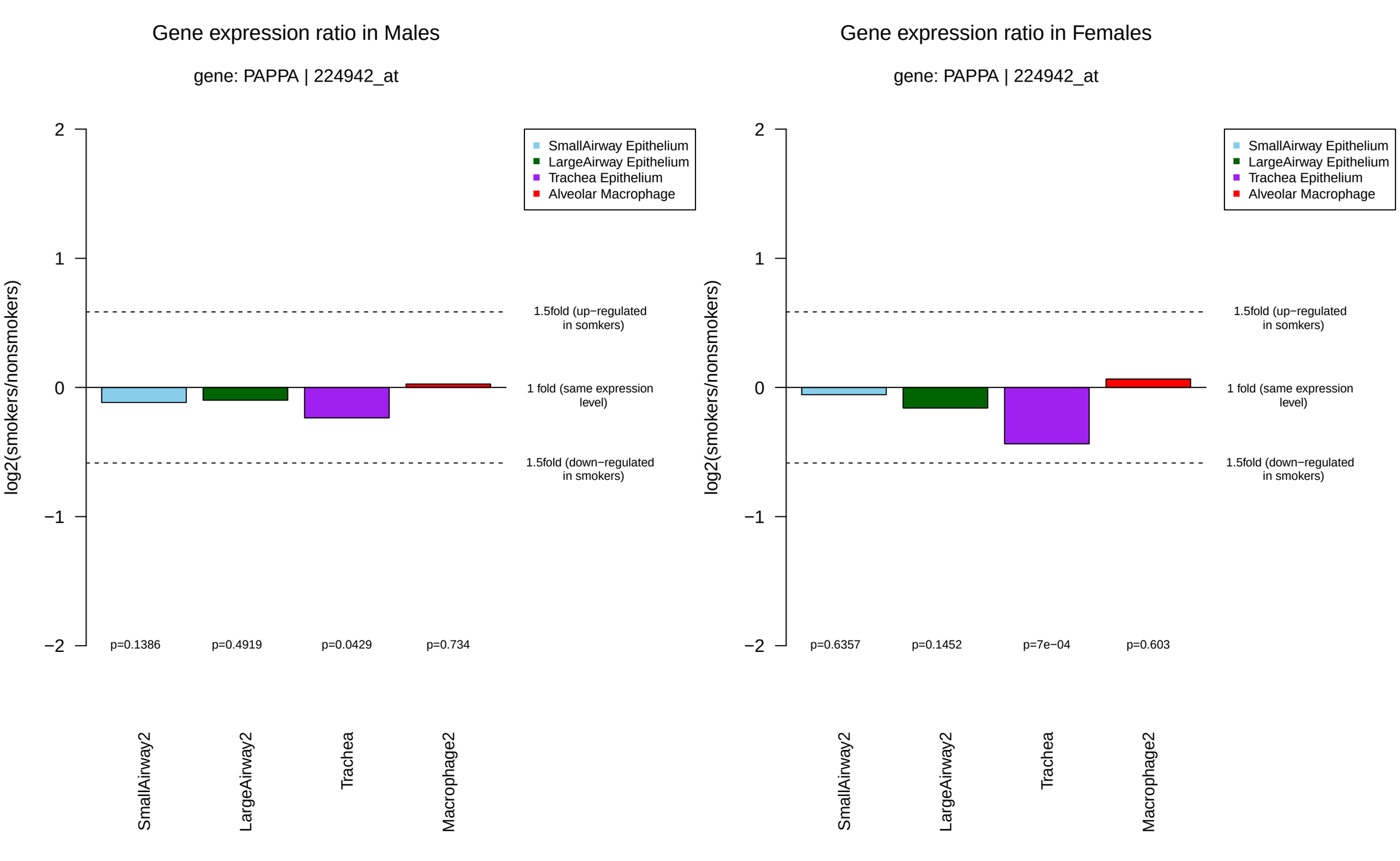
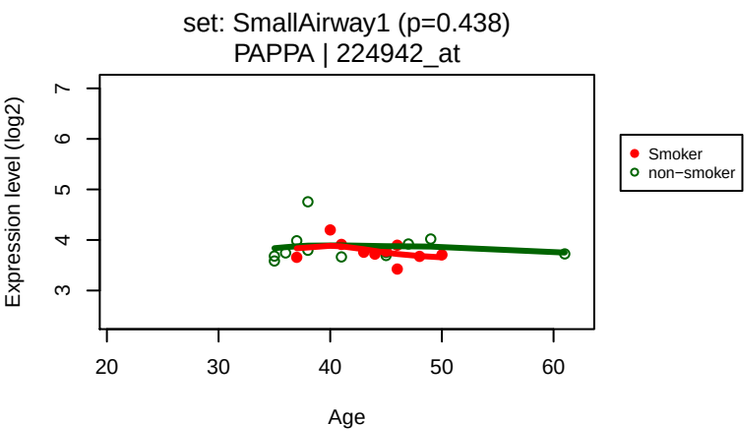
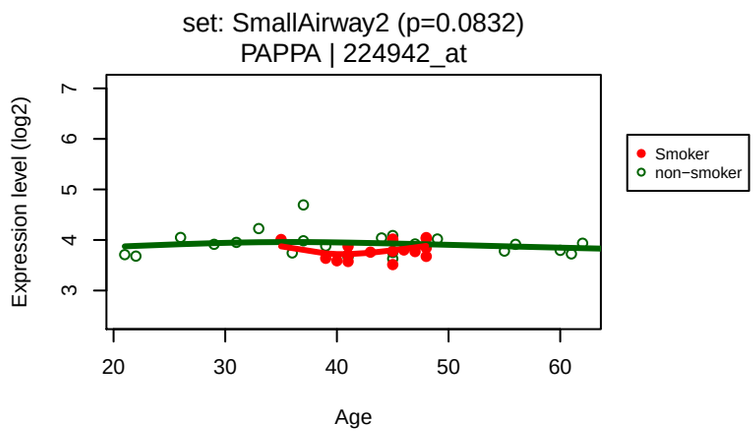
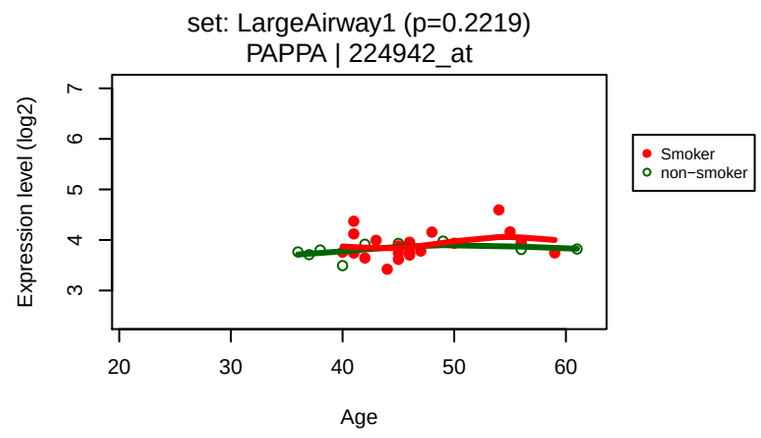

Gene expression of smokers and nonsmokers
gene: PAPPA | 224942_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.438
p=0.0832
p=0.21
p=0.2219
p=0.2305
p=0.0042
p=0.1034
p=0.5128
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

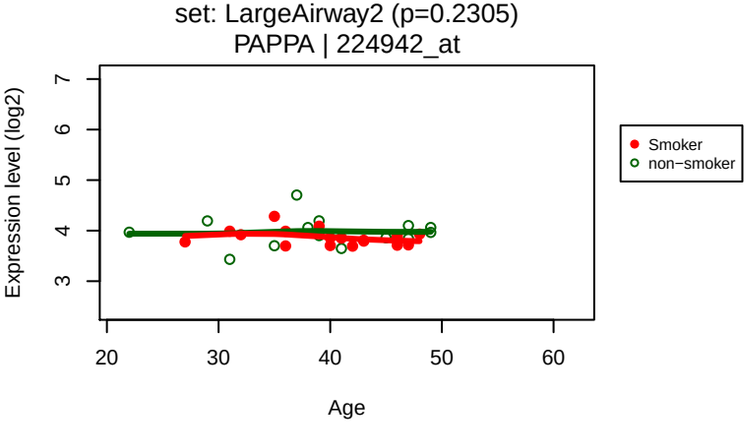

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 224942_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.438
p=0.0832
p=0.21
p=0.2219
p=0.2305
p=0.0042
p=0.1034
p=0.5128
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

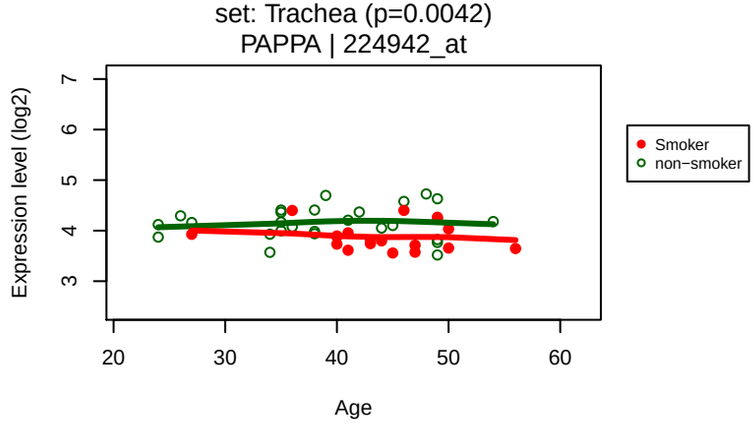

Gene expression ratio in Males
gene: PAPPA | 224942_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1386
p=0.4919
p=0.0429
p=0.734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 224942_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6357
p=0.1452
p=7e−04
p=0.603
SmallAirway2
LargeAirway2
Trachea
Macrophage2

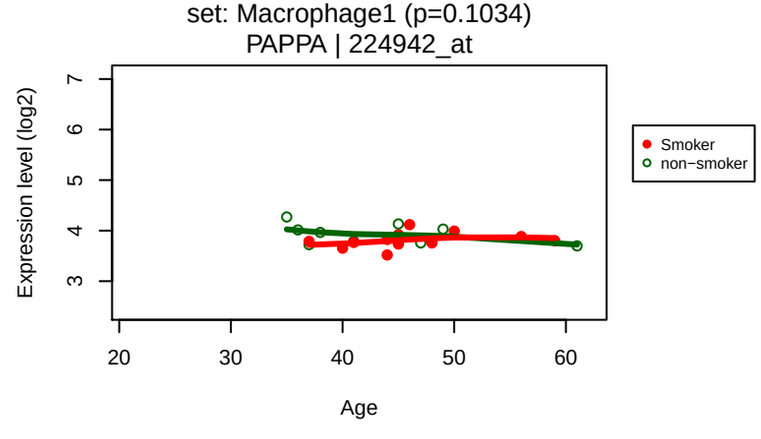

set: SmallAirway1 (p=0.438)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker

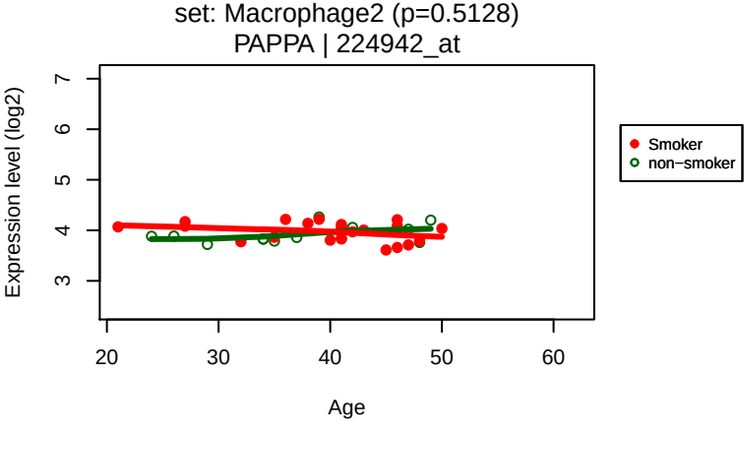

set: SmallAirway2 (p=0.0832)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker

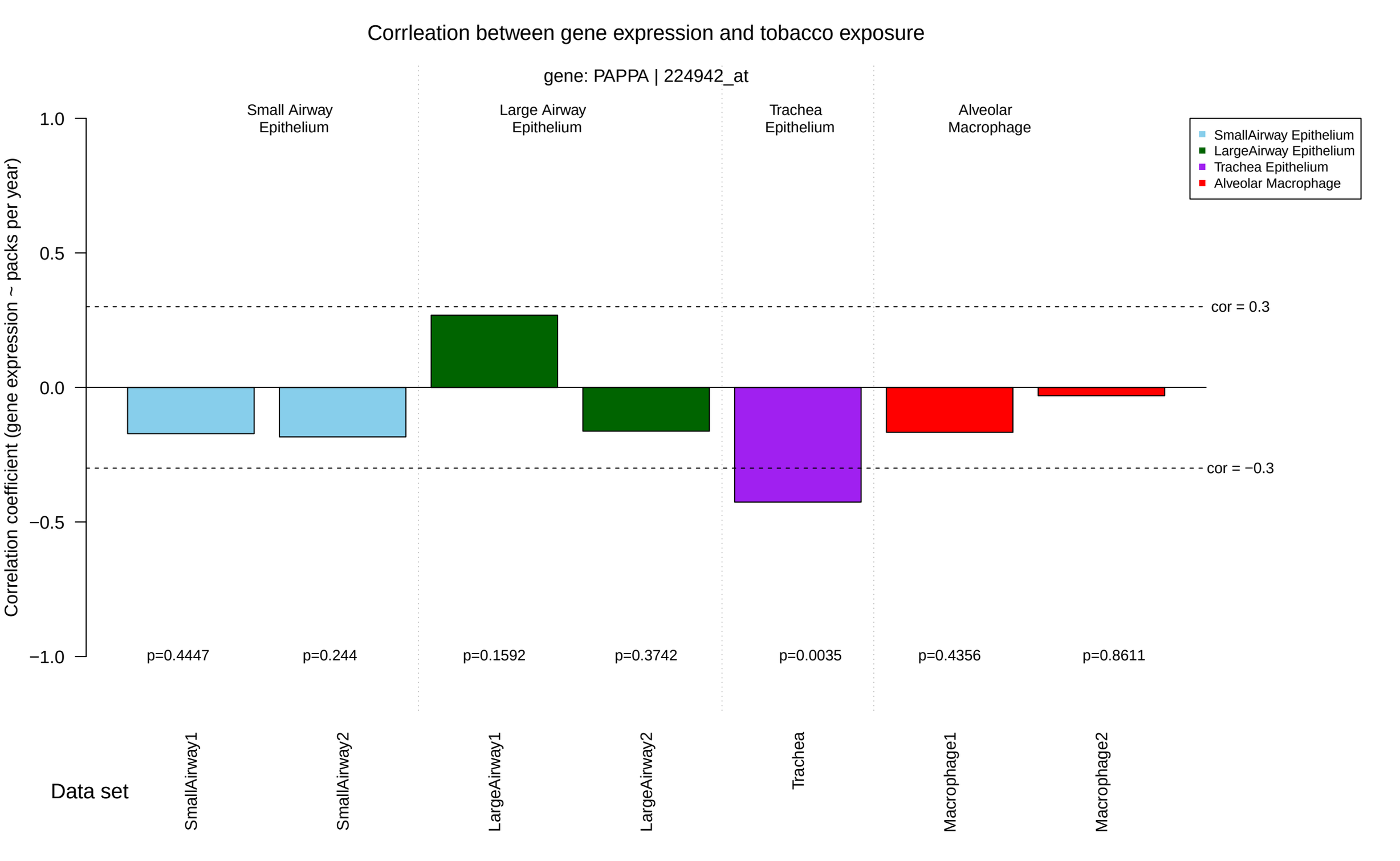

set: LargeAirway1 (p=0.2219)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2305)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1034)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5128)
PAPPA | 224942_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 224942_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4447
p=0.244
p=0.1592
p=0.3742
p=0.0035
p=0.4356
p=0.8611
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2